

Table A1: Output, replicability, and artifact-availability codes used to evaluate prior work on survey fraud.

Category	Code / Subcategory	Representative Papers
Primary Output Type	Tool or software system	[46, 62, 107]
	New algorithm or detection method	[10, 13, 23, 32, 38, 53, 55, 73, 78, 83, 93, 95, 101, 114, 117, 119, 126]
	Empirical measurement study	[1–3, 6, 7, 9, 11–13, 15–18, 24, 26–28, 31, 33, 35, 36, 40–43, 45, 52, 57, 58, 64–67, 69–72, 75, 77, 79, 81, 82, 86, 88, 90, 94, 96, 97, 103, 104, 108, 109, 111, 112, 116, 121, 125, 127]
	Guidelines or recommendations	[1, 3, 8, 9, 11, 12, 14, 20, 22, 24, 26, 28, 29, 31, 39, 41–43, 47, 48, 50, 52, 63, 64, 67, 69, 71, 74, 80, 82, 84, 92, 94, 97, 103–105, 108, 110, 111, 114–116, 118, 120, 121, 124, 127]
	Suggested/examples of questions	[1, 14, 25, 78, 86, 90]
Artifact Availability	Code provided	[2, 33, 45, 92, 106]
	Dataset provided	[2, 20, 33, 45, 92, 106, 115]

Table A2: The contribution and motivation of studies included in the SoK.

Category	Subcategory	Representative Papers
Contribution Type	User study	[1–4, 6, 7, 9–13, 16–18, 20, 23–30, 32, 33, 35, 36, 38–50, 52, 53, 55, 57–59, 62–76, 78–84, 86, 88–90, 92–97, 101–104, 106–112, 114–122, 124–127]
	Literature review	[8, 22, 34, 56, 61, 63, 80, 94, 121]
	Scoping review	[34, 37, 51, 99]
	Qualitative studies	[31, 60, 77, 85, 91, 98, 100, 105]
Study Motivation	Improve survey data quality	[1, 2, 7–9, 11, 12, 18, 22–24, 26–29, 31–35, 37–39, 41, 43, 44, 46–48, 52, 53, 55, 59–61, 63, 65–67, 69, 70, 73, 75, 76, 80, 82, 83, 86, 88, 90, 93, 94, 97, 109, 112, 114, 117–119, 121, 125, 127]
	Impacts of fraud on data quality	[3, 6, 16, 20, 40, 42, 45, 49, 56, 74, 77, 85, 89, 91, 103, 104, 111, 120]
	Excessive fraud in user study	[14, 17, 49, 57, 58, 62, 64, 71, 79, 81, 95, 98, 100–102, 105, 108, 115, 118, 122, 124, 126]
	Characterize fraudulent behavior	[4, 14, 15, 19, 36, 48, 65, 68, 69, 72, 84, 96, 113, 120]
	Propose a detection or mitigation strategy	[10, 13, 15, 23, 25, 29, 30, 34, 38, 39, 42, 45, 50, 51, 55, 56, 73, 78, 82, 92, 106, 108, 110, 114, 117, 119]
	Study the impact of AI-mediated fraud	[17, 40, 41, 43, 54, 57, 58, 62, 64, 71, 79, 81, 95, 99, 101, 107, 116, 123, 126]

Table A3: Fraud conceptualization codes: definitions, automation level, and types of fraudulent behavior.

Category	Code / Subcategory	Representative Papers
Definition Availability	Definition of Survey Fraud	[3, 6, 8, 10–12, 18, 20, 22–31, 33, 36, 41, 45–50, 53, 56–60, 63–66, 69, 71–73, 75, 77, 78, 81, 82, 84, 88, 94–96, 103, 104, 108, 114, 115, 117, 119]
	Definition of Careless Response	[1, 7, 9, 13, 15, 32, 34, 35, 37–39, 51, 52, 67, 70, 74, 80, 83, 86, 90, 93, 97, 106, 108, 109, 112, 121, 125]
Level of Automation	Traditional Bot	[2, 3, 12, 17, 29, 32, 33, 40, 41, 43, 45, 48, 49, 55–58, 60, 62, 64, 68, 71, 79, 81, 95, 101, 108, 111, 115, 125, 126]
	AI-based Bot	[32, 55, 66, 99, 108]
Type of Fraud	Location-based	
	Multiple Submissions	[42, 44, 47, 48, 50, 60, 84, 115]
	Inattentive	[1, 9, 30, 32, 35, 38, 67, 70, 80, 83, 86, 90, 93, 97, 109, 112, 121, 125]
Fraud Vector: Eligibility	Misreported demographics	[4, 98]
	Age misrepresentation	[49]
	Location spoofing (e.g., VPN)	[2–4, 108, 110, 118, 127]
	Identity spoofing / false claims	[2, 8, 11, 12, 18, 22–29, 31, 36, 41, 42, 45–48, 50, 53, 57–59, 63–65, 68, 69, 71–73, 75–78, 81, 82, 84, 88, 94–96, 103, 104, 114, 117, 119]
	Multi-accounting / duplicate submissions	[3, 8, 42, 44, 47, 48, 76, 78, 84, 92, 98, 108, 110, 118, 120, 124, 127]
Fraud Vector: Response-Integrity	Careless responding	[1–3, 7, 9, 10, 13, 15, 29, 30, 32, 34, 35, 37–39, 47, 52, 67, 70, 74, 76, 80, 83, 86, 90, 93, 97, 109, 110, 112, 115, 118, 120, 121, 124, 125]
	Mischievous responders	[3, 6, 16, 20, 45]
	Bot-generated	[2, 3, 12, 14, 17, 29, 32, 33, 40, 41, 43, 55, 57, 58, 62, 64, 66, 71, 79, 81, 92, 95, 101, 108, 110, 111, 115, 116, 125–127]

Table A4: Distribution of prior work on survey fraud across research domains.

Category	Subcategory	Count	Representative Papers
Research Domain	Health Sciences	40	[12, 16, 18, 20, 22, 24, 26, 28, 29, 31, 36, 40, 42, 44, 46, 48–50, 68, 69, 71, 73, 75, 77, 78, 81, 82, 85, 88, 91, 94, 98, 102–105, 114, 118, 119, 124]
	Psychology	28	[2, 17, 32, 33, 37, 45, 51–53, 55, 60, 61, 63, 70, 72, 74, 79, 83, 93, 97, 101, 106, 108, 111, 112, 121, 122, 125]
	Social Sciences	19	[1, 3, 9, 13, 15, 23, 27, 34, 39, 41, 47, 58, 62, 76, 84, 86, 116, 123, 126]
	Computer Science	19	
	• HCI	14	[10, 19, 30, 38, 54, 56, 64, 66, 89, 90, 95, 99, 107, 113]
	• Security	4	[8, 96, 120, 127]
	• Software Engineering	1	[25]
	Research Methods	16	[4, 6, 7, 11, 14, 35, 43, 65, 67, 80, 92, 100, 109, 110, 115, 117]
	Natural Sciences	2	[57, 59]

Table A5: Detection and mitigation strategy codes used to characterize prior work on survey fraud.

Category	Code / Subcategory	Representative Papers
Stage of Detection	Recruitment phase	[4, 8, 18, 22, 24, 26, 28, 30, 41, 44, 46, 63, 65, 68, 71, 82, 92, 94, 103, 105, 111, 119, 122]
	In-survey phase	[1–3, 6, 8, 9, 14, 15, 25, 29, 32, 35, 37, 38, 42, 45, 48, 50, 52, 53, 55, 62, 70, 72, 74, 78, 80, 83, 84, 86, 90, 92, 93, 95, 97, 101, 108–110, 112, 116, 118, 120–122, 127]
	Post-hoc	[3, 7, 8, 10–14, 16, 17, 20, 23, 27, 29, 31, 33, 37, 39, 40, 43, 45, 47, 48, 57–59, 64, 66, 67, 69, 73, 75–77, 79, 81, 88, 96, 104, 106, 115, 117, 118, 120, 124–127]
Metadata-based Methods	IP address / geolocation	[3, 18, 24, 28, 29, 41, 42, 45, 48, 50, 63, 65, 68, 71, 76, 82, 84, 92, 94, 105, 108, 110, 111, 115, 116, 118, 119, 127]
	VPN / proxy detection	[40, 48, 92, 118, 126, 127]
	Device / browser fingerprinting	[78, 95, 101, 108, 120, 127]
	Time	[3, 10, 13–15, 29, 30, 39, 42, 44, 45, 48, 50, 52, 67, 68, 74, 76, 84, 92, 102, 105–108, 110, 115, 116, 120, 122, 124, 125, 127]
	Server-side logs	[10, 27, 120, 127]
Response-based Methods	Attention checks / instructional checks	[1–3, 8, 9, 14, 15, 30, 45, 52, 62, 68, 70, 74, 80, 83, 84, 86, 90, 92, 97, 105–109, 112, 114, 116, 118, 120–122, 127]
	Error-count checks	[53, 74, 92, 105, 108, 115, 116, 127]
	Randomness filtering	[7, 32, 45, 76]
	Semantic similarity / embedding analysis	[48, 55, 68, 95, 116, 127]
	LLM detectors / AI-content detection	[38, 66, 107]
	Stylometry / writing-style analysis / internal consistency	[6, 14, 29, 33, 38, 39, 42, 45, 48, 52, 70, 72, 74, 84, 88, 92, 96, 98, 102, 105, 107, 108, 110, 115–118, 120, 122, 127]
	Honesty checks	[16, 17, 35, 39, 52, 74, 106, 108]
	Low-incidence questions	[52, 72, 74, 75, 92, 105, 108]
Platform-Level Mechanisms	Knowledge	[25, 59, 78]
	Quality controls / platform qualification filters	[22, 43–46, 57, 58, 64, 68, 73, 78, 81, 103, 105, 108, 115, 118, 120, 122, 127]
	Reputation scores / approval ratings	[69, 120, 122]
	CAPTCHA / ReCAPTCHA	[3, 29, 45, 48, 68, 84, 92, 102, 105, 107, 108, 110, 111, 115, 118, 127]
Manual Review	Manual review	[4, 11, 12, 14, 23, 29, 31, 44, 45, 47, 50, 68, 76–78, 92, 98, 104, 105, 107, 108, 110, 115, 118, 120, 122, 124, 127]
Detection Response	Prevent response completion	[105, 126, 127]
	Remove data	[14, 20, 29, 47, 48, 69, 76, 88, 98, 102, 108, 110, 115, 117, 118, 120, 122, 127]
	Deny compensation	[4, 29, 41, 45, 47]
	Track participant (internally)	[27, 48, 68]
	Report to platform	[47, 65]

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A Supplementary Material Availability

We make the full codebook and additional tables of representative papers on fraud conceptualization, prevention, and detection available as supplementary material here: <https://anonymous.4open.science/r/SoK-Mapping-Threats-to-Defenses-in-Online-Survey-Fraud-E9D9>.